GEOGRAPHY

PHYSICAL GEOGRAPHY

GEOMORPHOLOGY

MODULE - 18

VOLCANOES

- MECHANISM OF VULCANISM
- PLATE TECTONICS AND VULCANISM

GEOMORPHOLOGY

AS WE HAVE ALREADY SEEN…

WHAT GVR SIR HAS COVERED

- EARTH'S INTERIOR
- SOURCES OF HEAT IN EARTH'S INTERIOR
- CONTINENTAL DRIFT THEORY
- PLATE TECTONICS THEORY
- CONVERGENT, DIVERGENT & TRANSFORM PLATE BOUNDARIES

WHAT I WILL COVER

- VOLCANOES
- EARTHQUAKES
- GEOMORPHIC PROCESSES
- LANDFORMS & THEIR EVOLUTION
  1. FLUVIAL LANDFORMS
  2. GLACIAL LANDFORMS
  3. ARID LANDFORMS
  4. KARST TOPOGRAPHY
  5. COASTAL LANDFORMS

Quick relook of Module 17

VOLCANOES

VOLCANO AND VULCANISM

2 A volcano is a vent or opening nearly circular in form through which heated materials consisting of gases, liquid lava and fragments of rocks are ejected from the highly heated interior to the earth's surface.

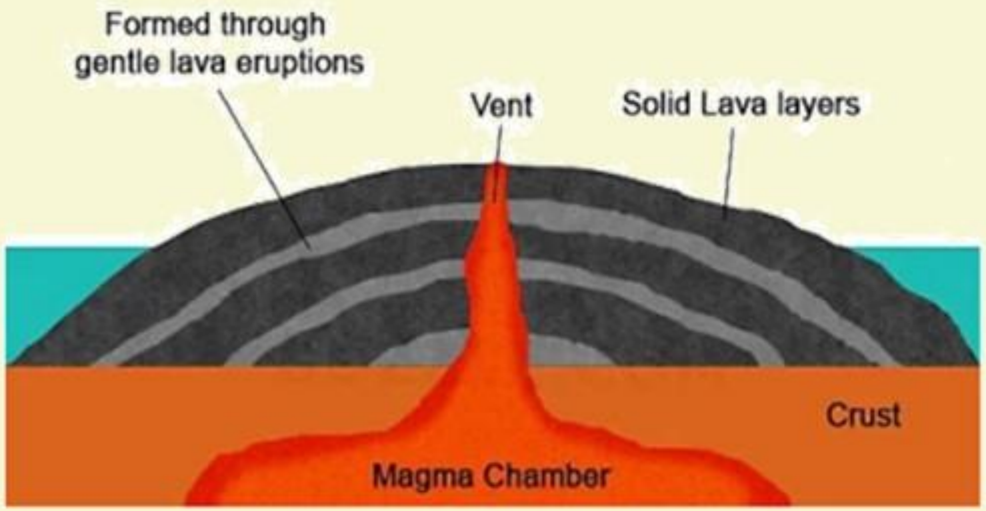
3 All phenomena connected with the movement heated material from the interior towards the surface of the earth is known as vulcanism.

Quick relook of Module 17

VOLCANOES

BASIC LAVAS

The Anatomy of a Shield Volcano



1. These are the hottest lavas, about 1,000°C and are highly fluid.
2. They are dark coloured like basalt, rich in iron and magnesium but poor in silica.
3. Once they come out of the vent, they flow quietly and they are not very explosive.
4. They flow readily with a speed of 10 to 30 miles per hour.
5. They affect extensive areas, spreading out as thin sheets over great distances before they solidify.
6. The resultant volcano is gently sloping with a wide diameter and forms a flattened shield or dome.

Learning Space

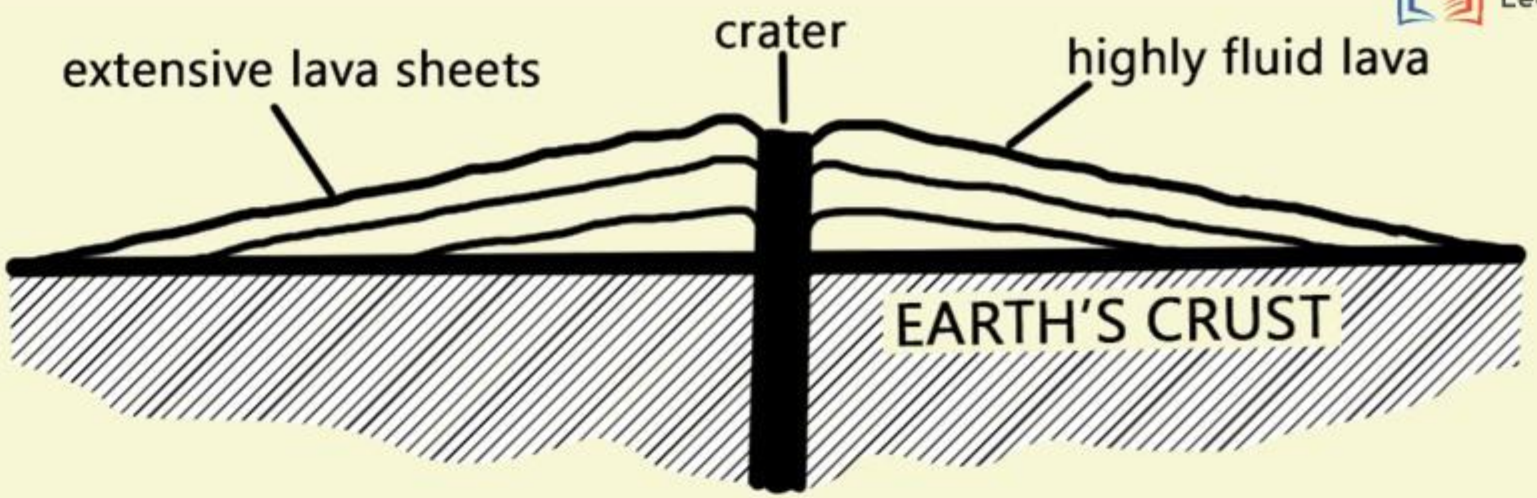
VOLCANOES

ACID LAVAS

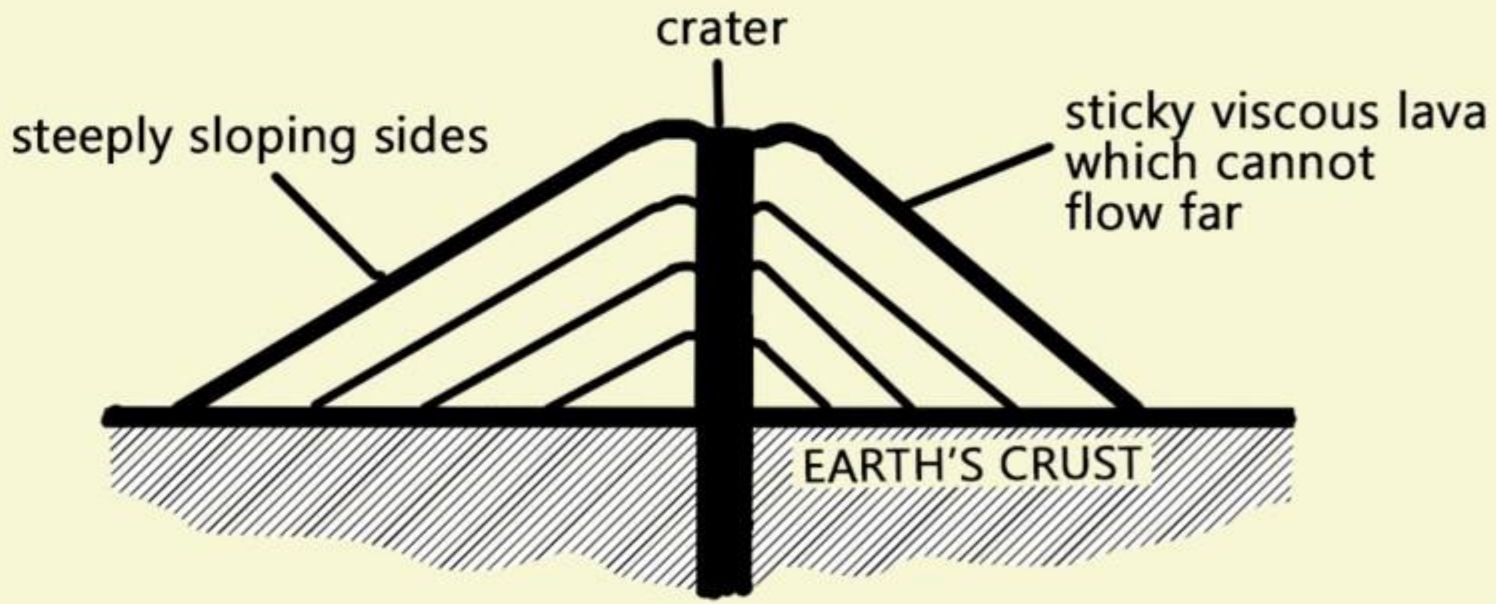
1. These lavas are highly viscous with high melting point.
2. They are light coloured, low density, high percentage of silica.
3. They flow slowly and seldom travel far before solidifying.
4. The resultant cone is steep-sided.
5. The rapid solidification of the lava in the vent obstruct the flow of the out-pouring lava that results in loud explosions, throwing out many volcanic bombs or pyroclasts.
6. Sometimes the lavas are so viscous that they form a spine or plug at the crater like that of Mount Pelee in Martinique.

Quick relook of Module 17

LAVA DOME OR SHIELD VOLCANO



ACID LAVA CONE



Quick relook of Module 17

CLASSIFICATION OF VOLCANOES

CLASSIFICATION ON BASIS OF MODE OF ERUPTION

CENTRAL ERUPTION TYPE OR EXPLOSIVE ERUPTION TYPE

- Hawaiian Type
- Strombolian Type
- Vulcanian Type
- Pelean Type
- Vesuvius Type

FISSURE ERUPTION TYPE

CLASSIFICATION ON THE BASIS OF PERIODICITY OF ERUPTION

- Active Volcanoes
- Dormant Volcanoes
- Extinct Volcanoes

Quick relook of Module 17

CLASSIFICATION OF VOLCANOES

1 CLASSIFICATION ON THE BASIS OF PERIODICITY OF ERUPTION

Mt Etna in Sicily

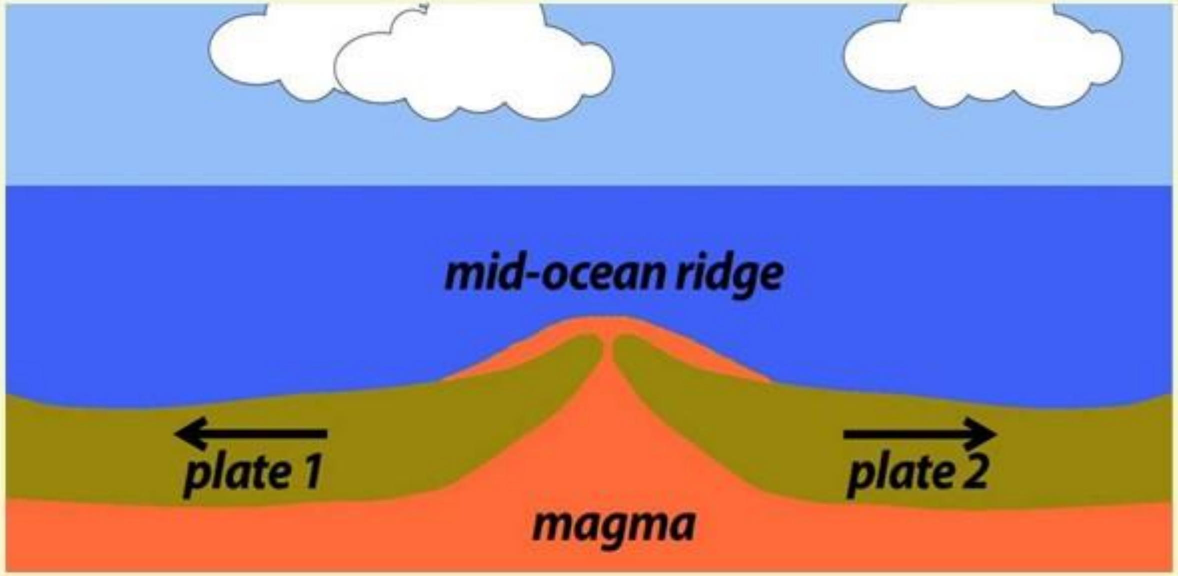
ACTIVE VOLCANOES

1. Active Volcanoes are those which constantly eject volcanic lavas, gases, ashes and fragmental materials.
2. Most of the active volcanoes are found along the mid oceanic ridges representing divergent plate margins (constructive plate margins) and in convergent plate margins (destructive plate margins).
3. Mount Etna and Mount Stromboli in the Mediterranean sea are the most significant examples of this category.

PLATE TECTONICS AND VULCANISM

MECHANISM AND CAUSES OF VULCANISM

WHAT HAPPENS AT CONSTRUCTIVE BOUNDARIES?



1. Theory of Plate Tectonics very well explains the mechanism of volcanism. In fact, volcanic eruptions are closely associated with the plate boundaries.

2. It may be pointed out that the type of plate movements and plate boundaries determines the nature and intensity of volcanic eruption.

3. Most of the active fissure volcanoes are found along the mid-oceanic ridges which represent splitting zones of divergent plate boundaries.

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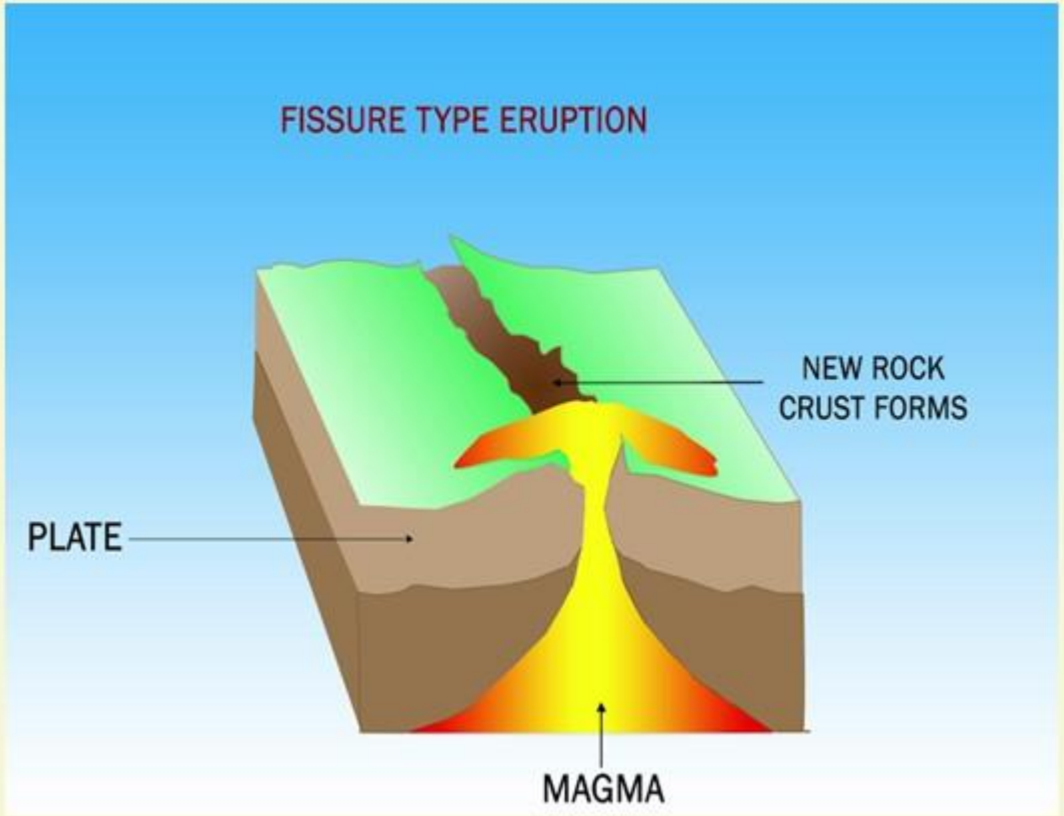

4. Two plates move in opposite direction from the mid-oceanic ridges due to thermal convective currents. This causes splitting and lateral spreading of plates and creates fractures and faults which causes pressure release.

5. The emergent magma moves upwards under the impact of accumulated gases and vapour.

6. This rise of magmas along the mid-oceanic ridges causes fissure type eruption.

PLATE TECTONICS AND VULCANISM

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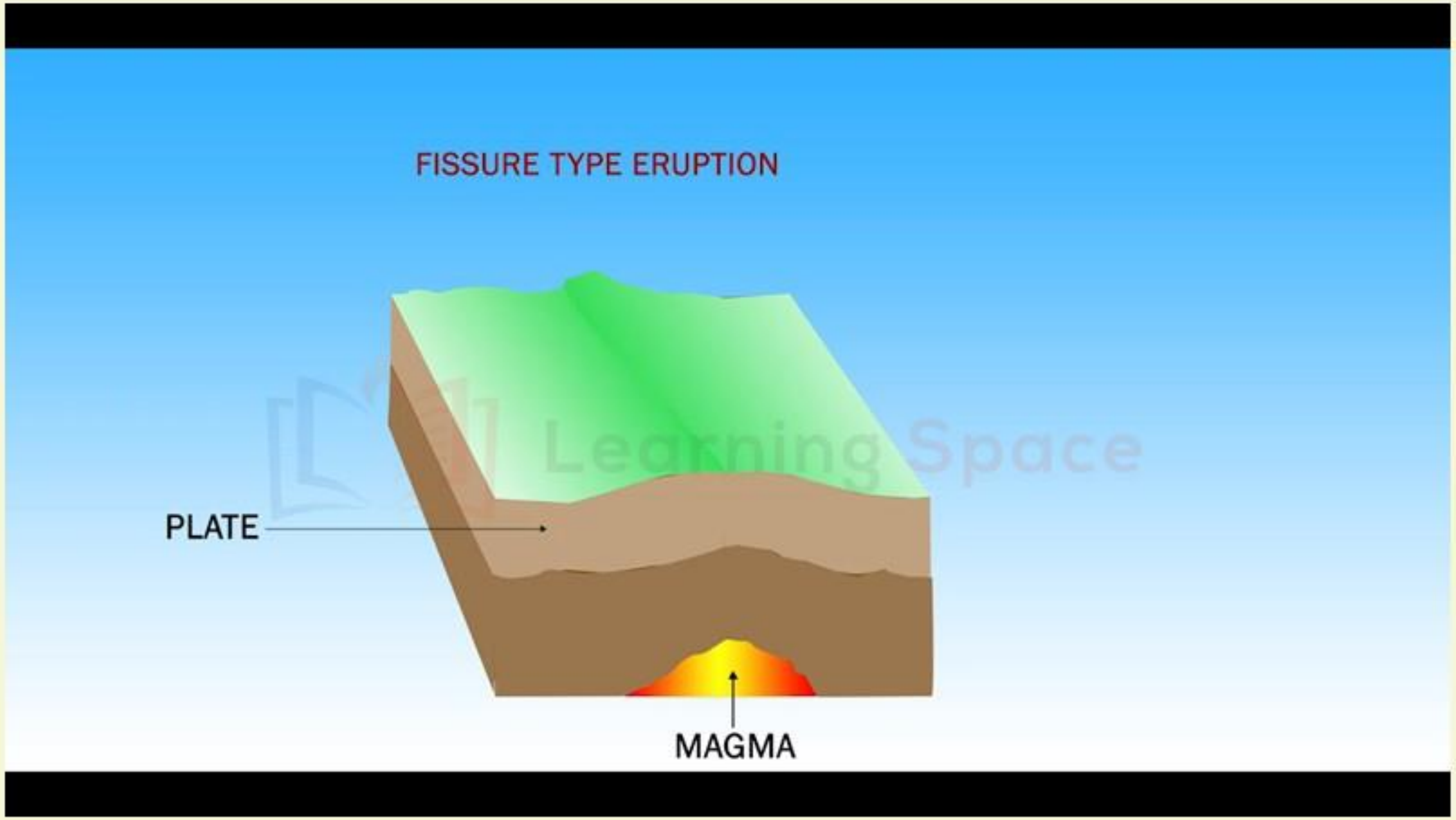
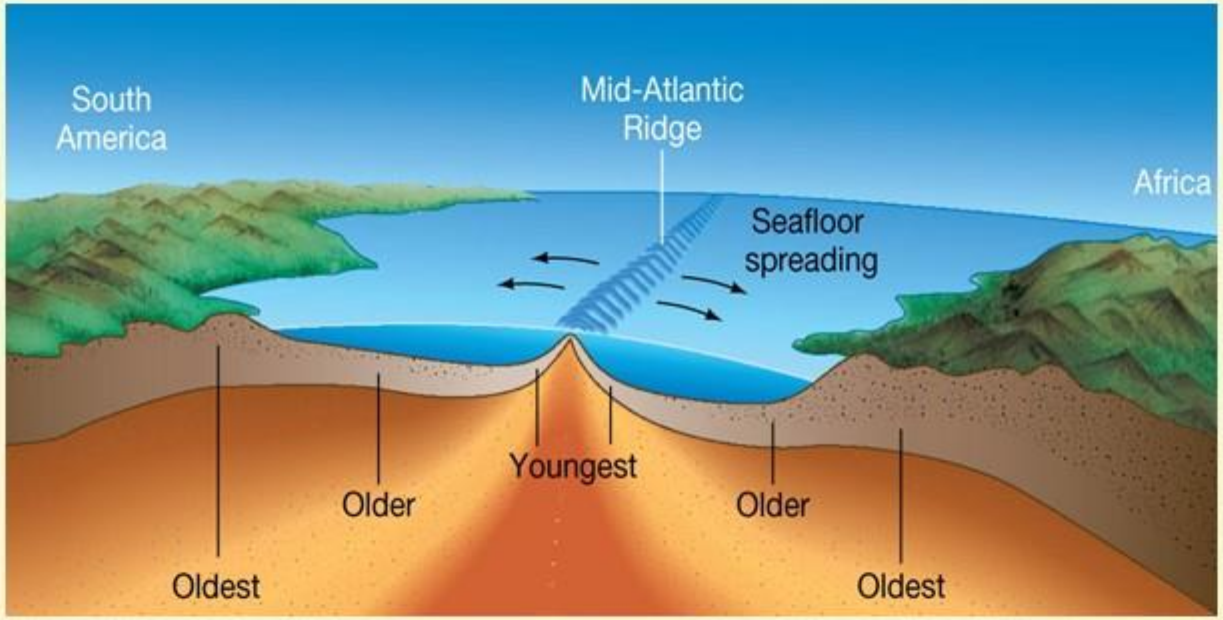


PLATE TECTONICS AND VULCANISM

MECHANISM AND CAUSES OF VULCANISM

WHAT HAPPENS AT CONSTRUCTIVE BOUNDARIES?

CONCEPT OF SEAFLOOR SPREADING



1. The constant upwelling of magma results in continuous cycles of cooling and solidification and thus there is constant creation of new basaltic crust. (Remember: This is the concept behind sea-floor spreading as well.)
2. Eg: Volcanic eruptions of Iceland and islands located along the mid-Atlantic ridges.
3. It is obvious that divergent or constructive plate boundaries are always associated with quiet fissure type eruptions because the pressure release of superincumbent load due to divergence of plates and formation of fissure and faults is a slow and gradual process.

PLATE TECTONICS AND VULCANISM

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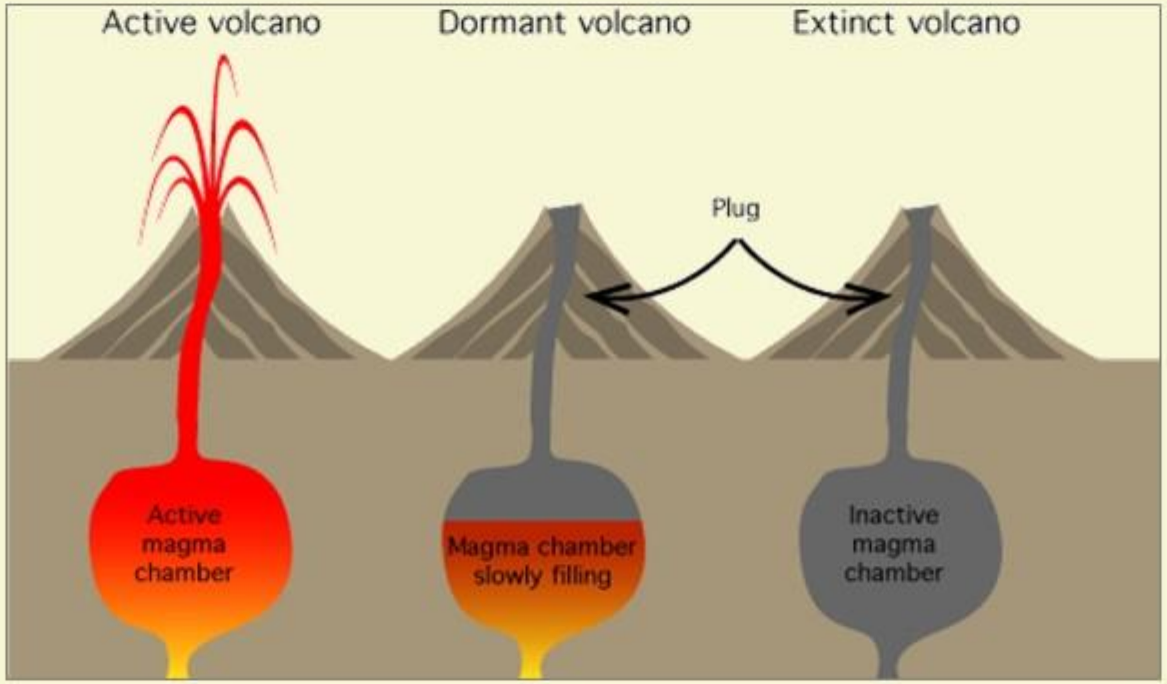

ACTIVENESS OF VOLCANOES

Since there is constant supply of basaltic lava from below the mid-oceanic ridges and hence the volcanoes are active near the ridges but the supply of lavas decreases with increasing distance from the mid-oceanic ridges and therefore the volcanoes become inactive, dormant and extinct depending on their distances from the source of lava supply.

PLATE TECTONICS AND VULCANISM

MECHANISM AND CAUSES OF VULCANISM

WHAT HAPPENS AT DESTRUCTIVE BOUNDARIES?

1. Destructive or convergent plate boundaries are associated with explosive type of volcanic eruptions.

2. When two convergent plates collide along the Benioff zone (subduction zone), comparatively heavier plate margin is subducted beneath lighter plate boundary.

3. The subducted plate margin, after reaching a depth of 100km or more in the upper mantle, gets melted and thus magma is formed.

4. This magma is forced to ascend by the enormous volume of accumulated explosive gases and thus magma appears as violent volcanic eruption on the earth's surface.

5. Such type of volcanic eruption is very common along the destructive plate boundaries which represent the volcanoes of the Circum-Pacific Belt and the Mid-Continental Belt.

6. The volcanoes of island arcs off the coast of Asia (Indonesia, Phillipines) are caused due to such subduction mechanism of oceanic crust.

PLATE TECTONICS AND VULCANISM

MECHANISM AND CAUSES OF VULCANISM

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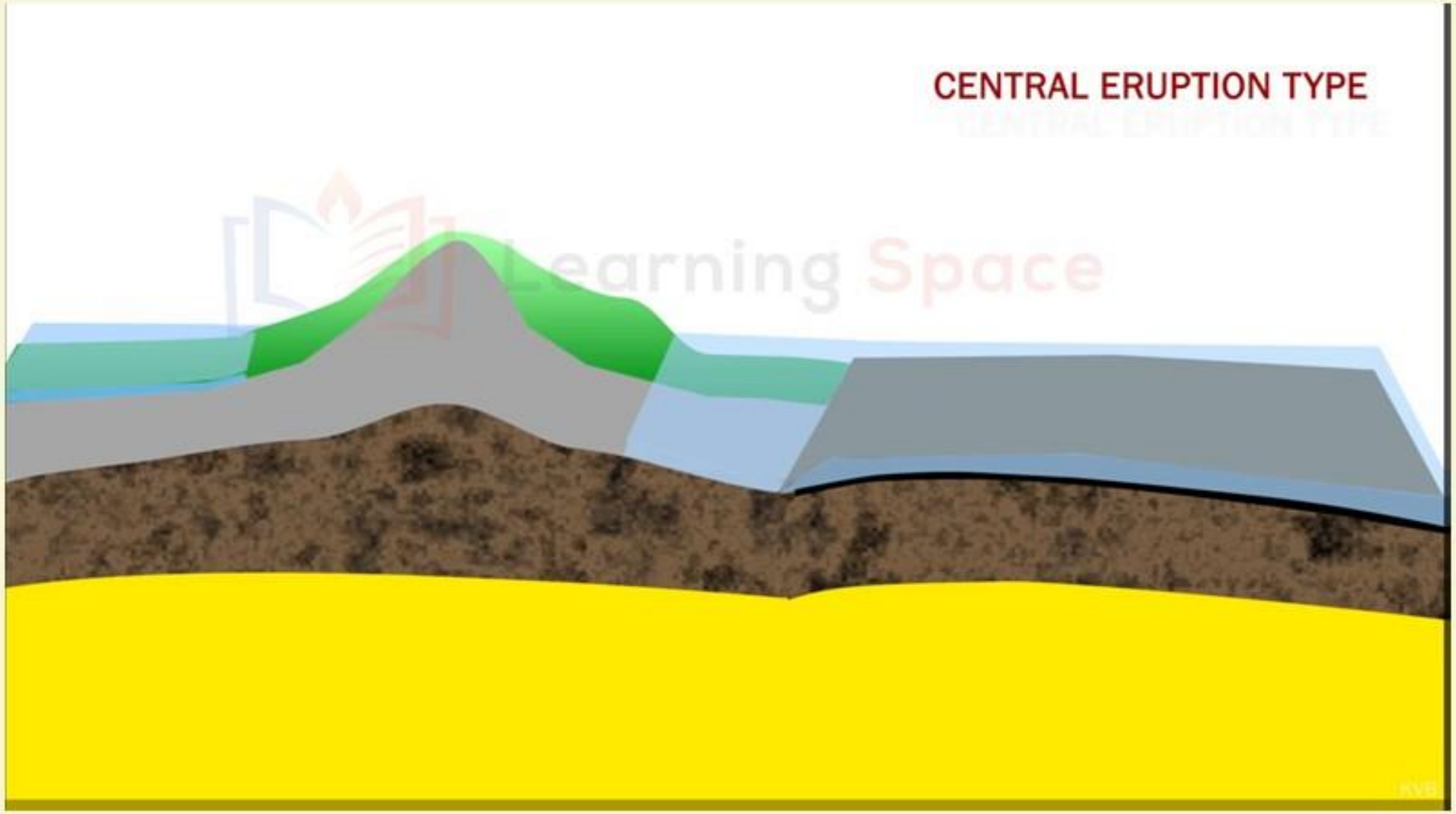

VOLCANOES

2014 GS-1 MAINS QUESTION

Topic: GS Paper 1 : Salient features of world's physical geography;

Important Geophysical phenomena such as earthquakes,

Tsunami, Volcanic activity, cyclone etc.,

2014 GS-1 paper had 100 marks of Geography question!

Q 1 Why are the world's fold mountain systems located along the margins of continents? Bring out the association between the global distribution of Fold Mountains and the earthquakes and volcanoes. **(10 Marks) (150 words)**

HINTS:

1. Link Plate Tectonics concept and Mountain Building or Orogenesis.
2. Think about convergent plate boundaries collision.
3. Volcanoes – At Himalayas or At Andes?

PLATE TECTONICS AND VULCANISM

MECHANISM AND CAUSES OF VULCANISM

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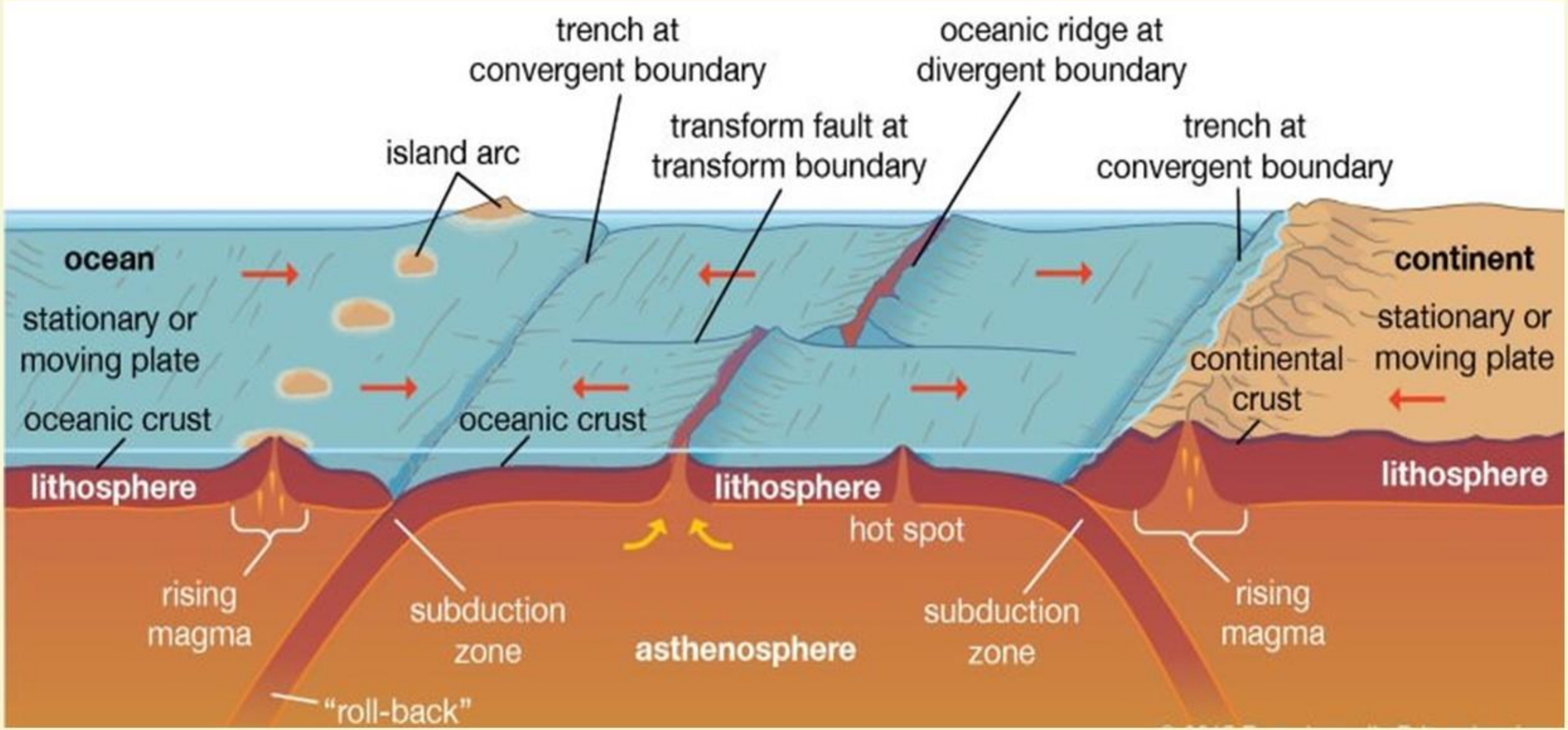
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oceanic crust
oceanic crust
lithosphere
lithosphere
lithosphere
hot spot
rising
magma
subduction
zone
asthenosphere
subduction
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magma
"roll-back"

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Q 2 Explain the formation of thousands of islands in Indonesian and Philippines archipelagos. **(10 Marks) (200 words)**

HINTS:

1. Ocean-ocean Convergence also known as island-arc convergence.
2. Also look at the "Mechanism of Vulcanism" part in this module.

PLATE TECTONICS AND VULCANISM

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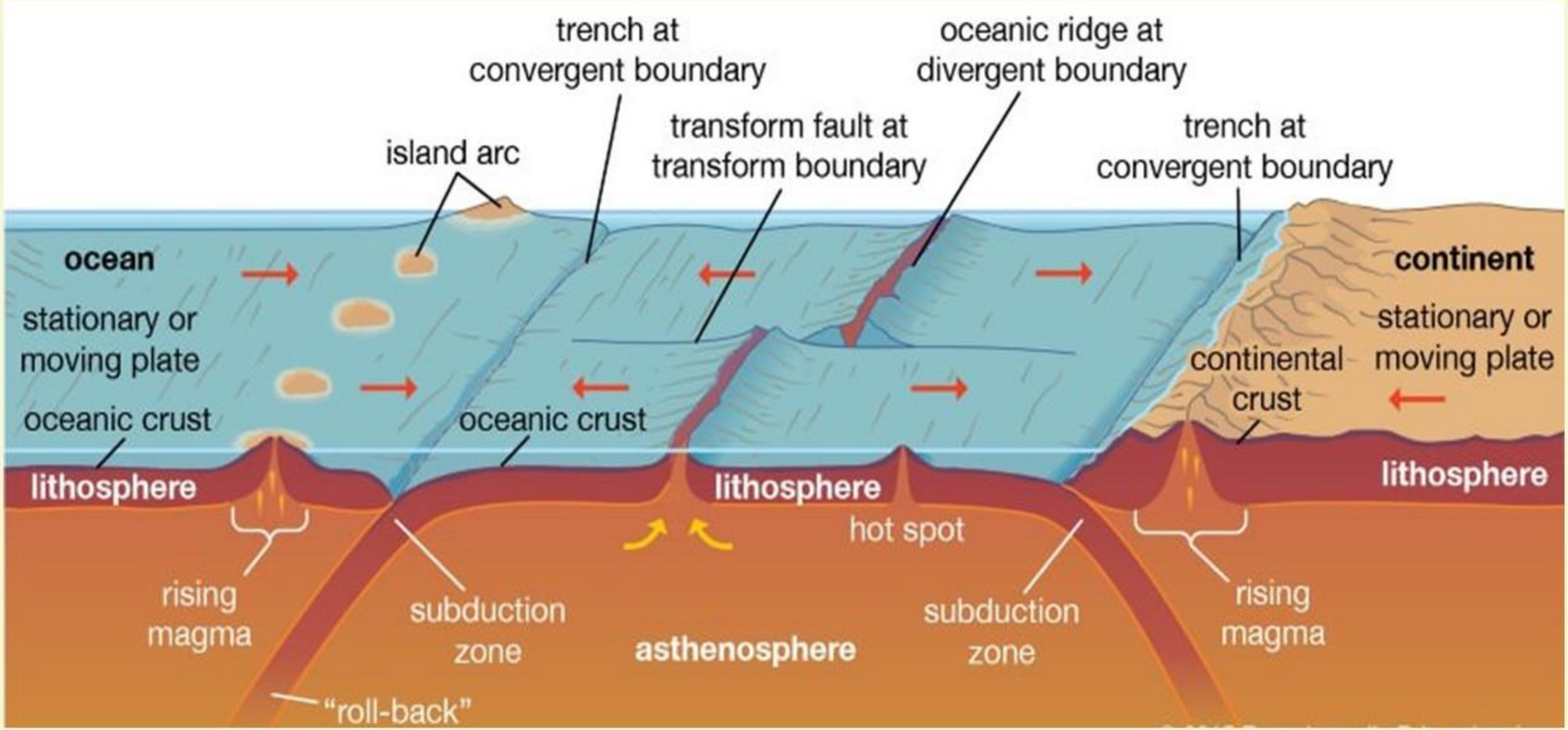
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